SIMATS ENGINEERING
ASSESSMENT TOOL3

COURSE CODE: MLA 02 **COURSE NAME: Fundamentals Of Machine Learning**

COURSE OUTCOME COVERED

CO2: *Analyze unsupervised learning methods, including clustering and dimensionality reduction, to extract insights from unlabelled data.(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 50 **Total Marks:50**

Annexure A

MCQ Test

Code of Conduct:

I ___________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Q1. Which clustering algorithm requires the number of clusters to be specified before training?
A) K-Means
B) Gaussian Mixture Model
C) DBSCAN
D) Hierarchical Clustering

Q2. What is the primary objective of the K-Means algorithm?
A) Maximize inter-cluster variance
B) Minimize the sum of squared distances between data points and cluster centroids
C) Classify data using labels
D) Maximize the number of clusters

Q3. Which method is commonly used to determine the optimal number of clusters in K-Means?
A) Gradient Descent
B) Elbow Method
C) Cross Validation
D) Grid Search

Q4. In K-Means clustering, a centroid represents:
A) The farthest data point in a cluster
B) The average position of all data points in a cluster
C) A randomly selected feature
D) The largest data value

Q5. Which distance measure is most commonly used in the standard K-Means algorithm?
A) Manhattan Distance
B) Euclidean Distance
C) Cosine Similarity
D) Hamming Distance

Q6. Which algorithm performs soft clustering by assigning probabilities to cluster memberships?
A) K-Means
B) Gaussian Mixture Model (GMM)
C) DBSCAN
D) K-Nearest Neighbors

Q7. The Expectation-Maximization (EM) algorithm consists of which two main steps?
A) Training and Testing
B) Selection and Mutation
C) Expectation and Maximization
D) Encoding and Decoding

Q8. During the Expectation (E) step, the EM algorithm primarily:

A) Updates model parameters
B) Calculates cluster membership probabilities
C) Removes noisy features
D) Splits the dataset

Q9. During the Maximization (M) step, the EM algorithm primarily:
A) Updates the model parameters
B) Calculates Euclidean distance
C) Reduces dimensions
D) Selects important features

Q10. Gaussian Mixture Models assume that data is generated from:
A) A single Gaussian distribution
B) Multiple Gaussian distributions
C) A decision tree model
D) A regression equation

Q11. Which clustering technique can model clusters with different shapes and sizes more effectively than K-Means?
A) K-Means
B) Gaussian Mixture Model
C) Linear Regression
D) Logistic Regression

Q12. What does the covariance matrix represent in a Gaussian Mixture Model?
A) Feature names
B) Variability and relationship among variables
C) Number of clusters
D) Missing values

Q13. The Curse of Dimensionality mainly occurs when:
A) The dataset has too many rows
B) The dataset has a large number of features
C) The dataset contains categorical data
D) The dataset has missing values

Q14. One major effect of the Curse of Dimensionality is:
A) Faster model training
B) Increased computational complexity
C) Reduced storage requirements
D) Improved prediction accuracy

Q15. Which technique is commonly used to address the Curse of Dimensionality?
A) Feature Scaling
B) Dimensionality Reduction
C) Data Duplication
D) Oversampling

Q16. Dimensionality reduction helps by:

A) Increasing the number of features
B) Removing redundant information
C) Creating duplicate records
D) Increasing model complexity

Q17. Principal Component Analysis (PCA) is primarily used for:
A) Classification
B) Dimensionality Reduction
C) Clustering only
D) Regression

Q18. PCA transforms the original variables into:
A) Decision Trees
B) Principal Components
C) Hidden Layers
D) Support Vectors

Q19. Principal components in PCA are:
A) Dependent on each other
B) Orthogonal to each other
C) Randomly selected
D) Binary variables

Q20. The first principal component captures:
A) The minimum variance
B) The maximum variance in the dataset
C) Only categorical features
D) The class labels

Q21. Which matrix is used in PCA to compute principal components?
A) Identity Matrix
B) Covariance Matrix
C) Sparse Matrix
D) Confusion Matrix

Q22. Eigenvalues in PCA indicate:
A) Number of observations
B) Amount of variance explained by each component
C) Cluster labels
D) Feature names

Q23. Eigenvectors in PCA represent:
A) Principal component directions
B) Class labels
C) Cluster centroids
D) Hidden variables

Q24. PCA is an example of:
A) Supervised Learning
B) Unsupervised Learning

C) Reinforcement Learning
D) Semi-supervised Learning

Q25. Which preprocessing step is generally recommended before applying PCA?
A) Data Standardization
B) Label Encoding
C) Feature Duplication
D) Image Augmentation

Q26. Factor Analysis primarily aims to identify:
A) Hidden (latent) variables
B) Decision boundaries
C) Class labels
D) Outliers

Q27. Factor Analysis assumes that observed variables are influenced by:
A) Random forests
B) Latent factors
C) Neural networks
D) Clusters only

Q28. Which technique is mainly used for identifying latent constructs in data?
A) PCA
B) Factor Analysis
C) Linear Regression
D) KNN

Q29. Independent Component Analysis (ICA) attempts to find:
A) Correlated components
B) Statistically independent components
C) Principal components only
D) Decision boundaries

Q30. ICA is widely used in:
A) Image Compression
B) Blind Source Separation
C) Linear Regression
D) Data Sorting

Q31. Which of the following assumes that source signals are statistically independent?
A) PCA
B) ICA
C) K-Means
D) Logistic Regression

Q32. Probabilistic PCA differs from conventional PCA because it:
A) Uses a probabilistic model
B) Requires labeled data

C) Uses decision trees
D) Eliminates covariance

Q33. Probabilistic PCA is especially useful when handling:
A) Missing or noisy data
B) Text classification
C) Image segmentation only
D) Database indexing

Q34. Which dimensionality reduction technique maximizes variance?
A) PCA
B) ICA
C) Factor Analysis
D) K-Means

Q35. Which technique emphasizes statistical independence rather than variance?
A) PCA
B) ICA
C) GMM
D) K-Means

Q36. Which clustering algorithm assigns each data point to exactly one cluster?
A) K-Means
B) Gaussian Mixture Model
C) Fuzzy C-Means
D) EM Algorithm

Q37. In Gaussian Mixture Models, cluster membership is represented by:
A) Binary values
B) Probability values
C) Decision rules
D) Distance only

Q38. Which algorithm is iterative and converges when parameter updates become negligible?
A) EM Algorithm
B) Breadth-First Search
C) Linear Search
D) Merge Sort

Q39. Which of the following is NOT a dimensionality reduction technique?
A) PCA
B) ICA
C) Factor Analysis
D) K-Means

Q40. The primary benefit of dimensionality reduction is:

A) Increased feature redundancy
B) Reduced computational cost
C) Increased class labels
D) Increased memory usage

Q41. PCA reduces dimensions while preserving maximum:
A) Correlation
B) Variance
C) Distance
D) Entropy

Q42. Which algorithm updates cluster centroids repeatedly until convergence?
A) K-Means
B) PCA
C) ICA
D) Decision Tree

Q43. Which measure is minimized in K-Means clustering?
A) Within-cluster Sum of Squares (WCSS)
B) Entropy
C) Information Gain
D) Gini Index

Q44. Which technique is most suitable for separating mixed audio signals?
A) PCA
B) ICA
C) K-Means
D) Naïve Bayes

Q45. Which method transforms correlated variables into uncorrelated components?
A) PCA
B) EM Algorithm
C) GMM
D) Factor Analysis

Q46. Which of the following is a probabilistic clustering algorithm?
A) Gaussian Mixture Model
B) K-Means
C) Decision Tree
D) Random Forest

Q47. Which technique is commonly used before visualization of high-dimensional data?
A) PCA
B) Linear Regression
C) Logistic Regression
D) Naïve Bayes

Q48. The main objective of Factor Analysis is to:

A) Reduce variables by identifying latent factors
B) Predict target values
C) Build classification models
D) Sort datasets

Q49. Which algorithm can assign a data point to multiple clusters with different probabilities?
A) Gaussian Mixture Model
B) K-Means
C) KNN
D) Decision Tree

Q50. Which of the following is an unsupervised machine learning technique?
A) K-Means Clustering
B) Logistic Regression
C) Decision Tree Classification
D) Support Vector Machine

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time

Question Interpretation	**10%**	**Interprets all questions correctly and responds appropriately**	**Misinterprets a few questions**	**Often misinterprets questions**	**Unable to understand questions**